How many jurors is a panel worth?

Condorcet jury theory and rater equivalence

Four datasets, six principal figures, and a reproducible comparison of observed panel accuracy with explicitly hypothetical independent jurors.

Dataset	Items	Panel	Single	Observed panel	iid model	iid jury needed
NBA forecasts	465	10	63.09%	65.89%	79.45%	3
Bluebirds	108	39	63.56%	75.93%	95.88%	7
City comparisons	1000	97	67.72%	76.02%	99.99%	5
Ducks	240	40	67.26%	68.75%	98.74%	3

The last column is the smallest whole model jury matching the observed panel at the displayed size, using that dataset's single-rater accuracy. It is a model projection, not a recovered count of independent real people.

Main interpretation. More recorded votes can improve accuracy, but the simple homogeneous independence model can substantially overpredict those gains. This gap does not identify a causal effect of dependence: common item difficulty, heterogeneous skill and finite-pool sampling also matter.

Scope. Majority curves use exact sampling without replacement. NBA uncertainty resamples 11 whole waves; bird uncertainty resamples images within truth classes. City results are descriptive because questions share city identities. All results condition on the observed people.

Empirical substitutions: fewer people, same target

These comparisons use real reference panels. They are distinct from the hypothetical iid equivalents on the previous page.

Evaluation set	Method	Reference	Whole people	95% interval	Fractional
Bluebirds	Training-selected pool	9 random	1	1 to 1	<=1
Bluebirds	Trained integer weights	9 random	3	2 to 4	2.33
City comparisons	Training-selected pool	9 random	5	Not estimated	4.26
City comparisons	Trained integer weights	9 random	7	Not estimated	6.42

Bluebirds: 54 training and 54 test images. Cities: 200 training and 215 test pairs with disjoint city identities; 198 incomplete pairs and 387 cross-set pairs excluded. The selected pool contains the top half of observed raters by training accuracy. One selected rater means an average over that pool, not one specific expert.

Bird intervals resample test images conditional on fixed training and workers. City counts are descriptive for the test pairs. These headcounts do not include the cost of acquiring training labels, and this single split does not establish universal savings.

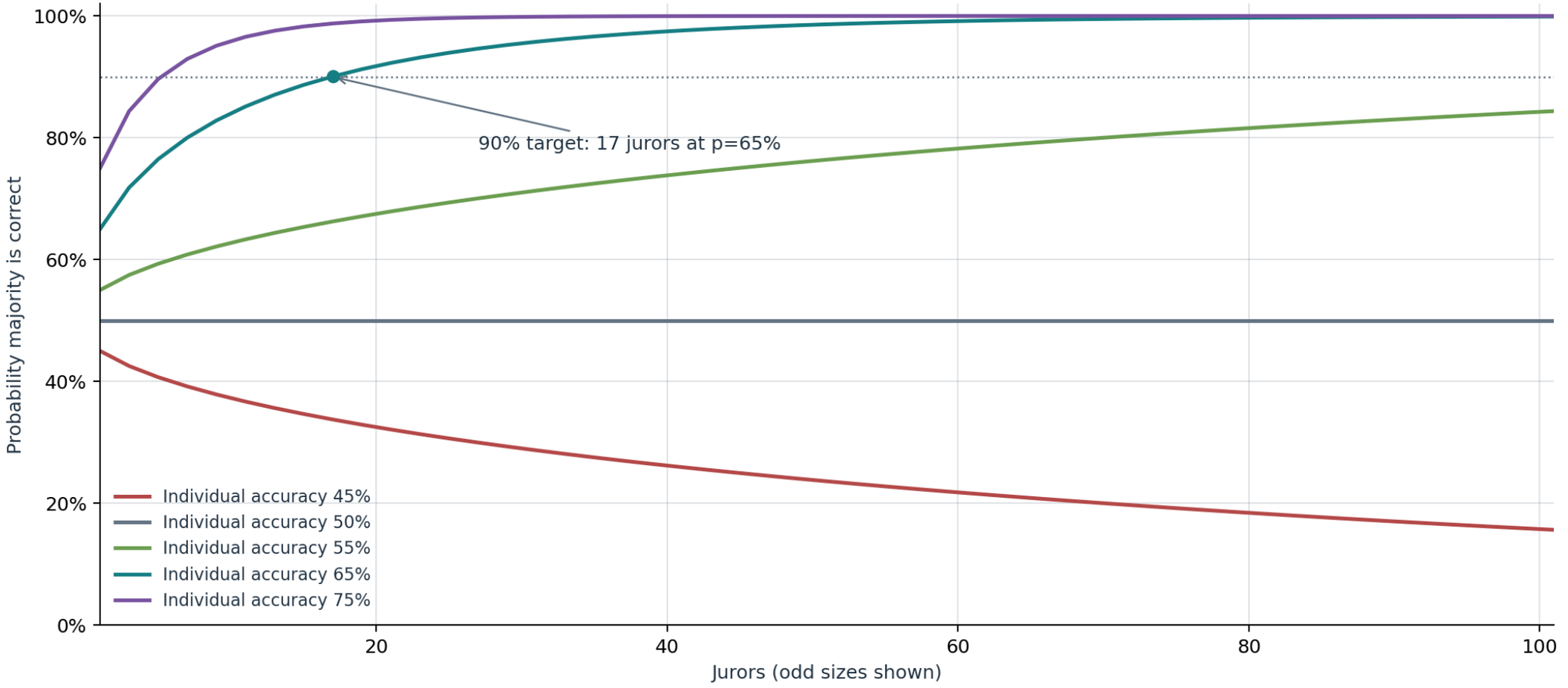
NBA: richer answers do not establish a saving

Ten winner-only respondents score 65.89%. Winner-only majority first reaches this score at nine people because of the odd/even tie plateau. Confidence weighting first reaches it at ten, and peer-agreement aggregation at nine. Their k=10 improvements are +0.14 and +0.43 percentage points, respectively; both paired intervals include zero.

For confidence weighting, 26.13% of bootstrap draws do not cross within k=1..10; for peer agreement, 21.16% do not cross. Those draws are retained as bounds. Fractional first crossings are 9.22 and 8.62; these should not be mistaken for whole recruited people.

1. What Condorcet predicts

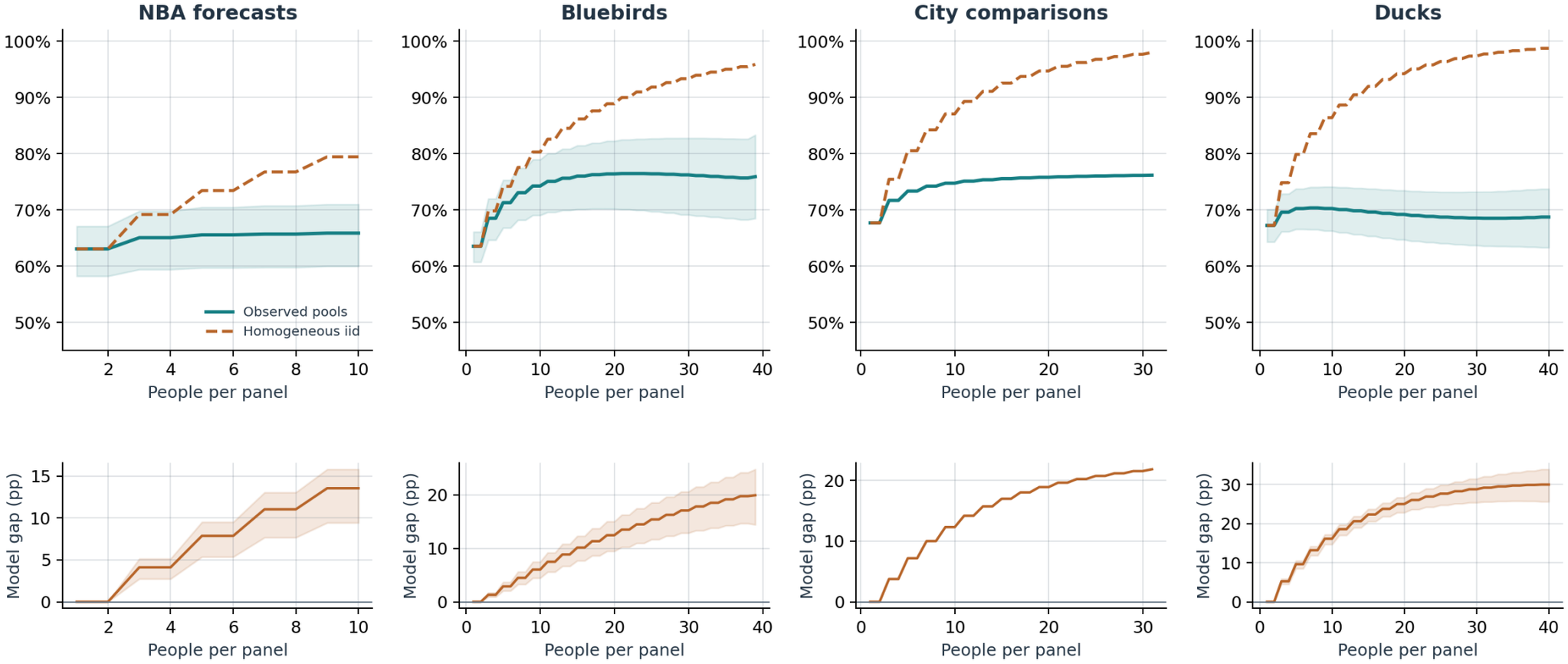
THEORETICAL | Independent, equally competent jurors; binary decisions; majority rule.



Analytical binomial tails, not recorded responses. At p=50% the curve stays at chance; below 50%, larger majorities worsen.
Source: classical Condorcet model; formulas and assumptions in the accompanying methods.

2. Real panels gain less than the simple model predicts

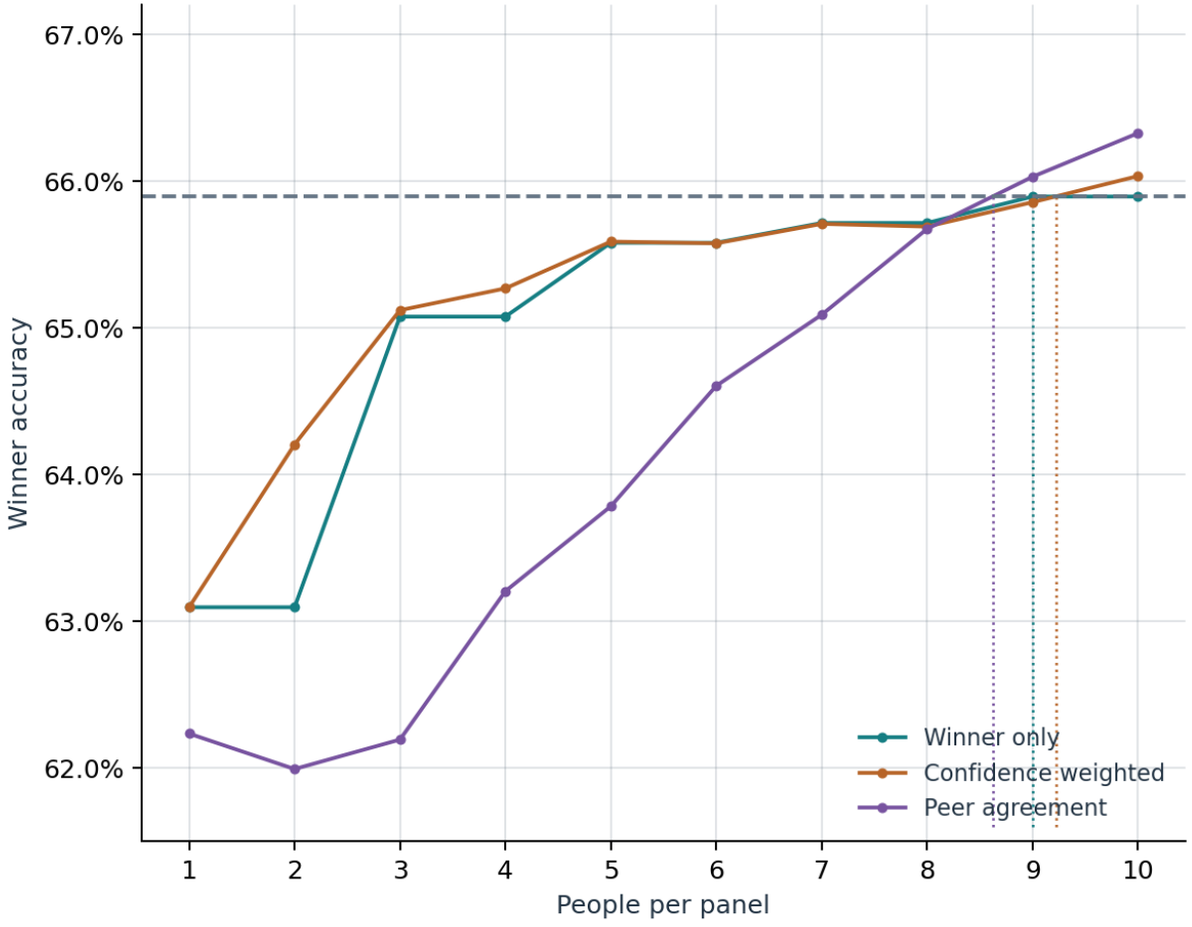
RECORDED RESPONSES + MODEL | Solid: exact observed-pool majority accuracy. Dashed: iid model matched to single-rater accuracy.
Shading: pointwise 95% resampling intervals where supported; the city curve is descriptive.



NBA: 11-wave bootstrap. Birds: truth-stratified image bootstrap. All intervals condition on the recorded people.
City view stops at k=31; all 1,000 items support k=97 (appendix). Gaps can reflect task difficulty, voter differences and sampling, not only dependence.

3. Can richer answers replace additional people?

NBA RECORDED RESPONSES | Match the accuracy of ten winner-only respondents on the same 465 games.



Reference accuracy: 65.89%

Winner only
Whole people: 9 | Interpolated: 9.0
95% whole-count interval: 5 to 9

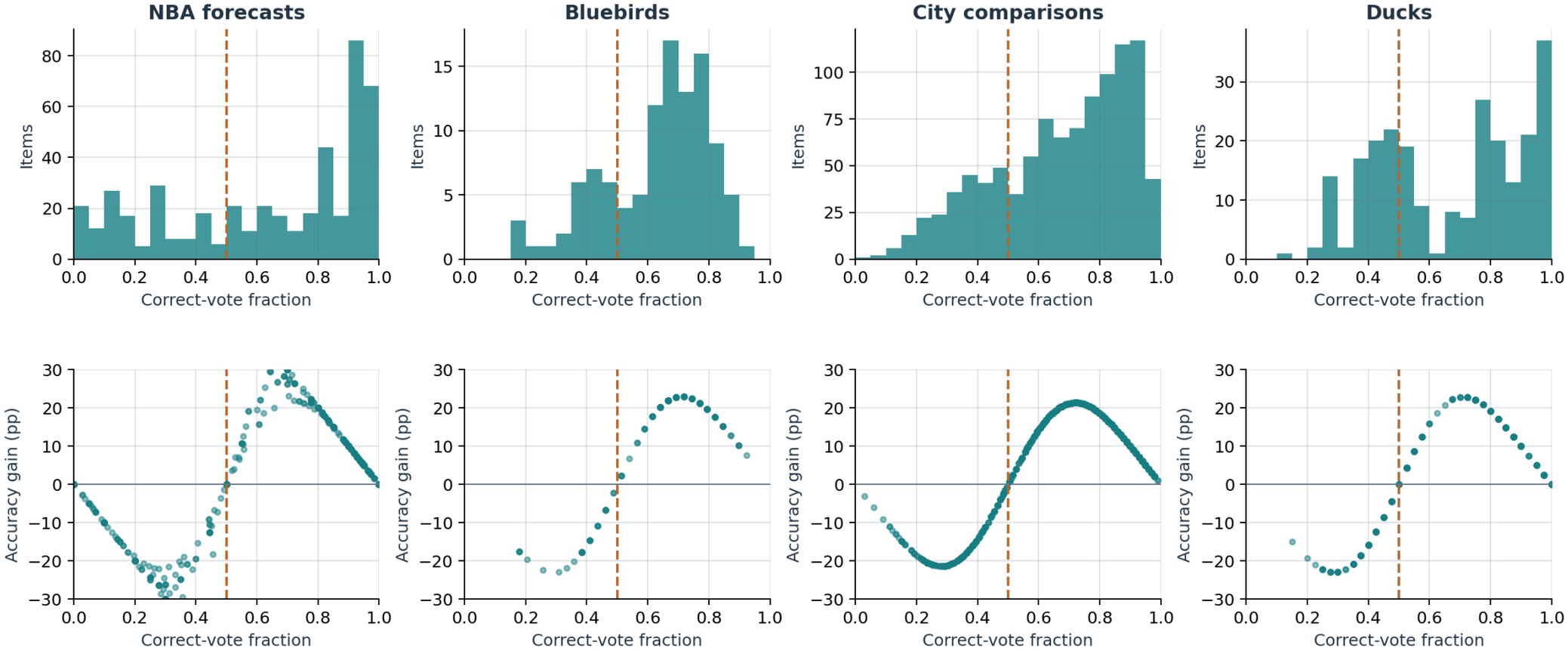
Confidence weighted
Whole people: 10 | Interpolated: 9.22
95% whole-count interval: 3 to >10

Peer agreement
Whole people: 9 | Interpolated: 8.62
95% whole-count interval: 7 to >10

Paired whole-wave bootstrap recomputes the target and each crossing; >10 draws are retained.
All three formats were recorded together. At k=10, both richer-method improvement intervals include zero; no established respondent saving.

4. Which questions defeat the crowd?

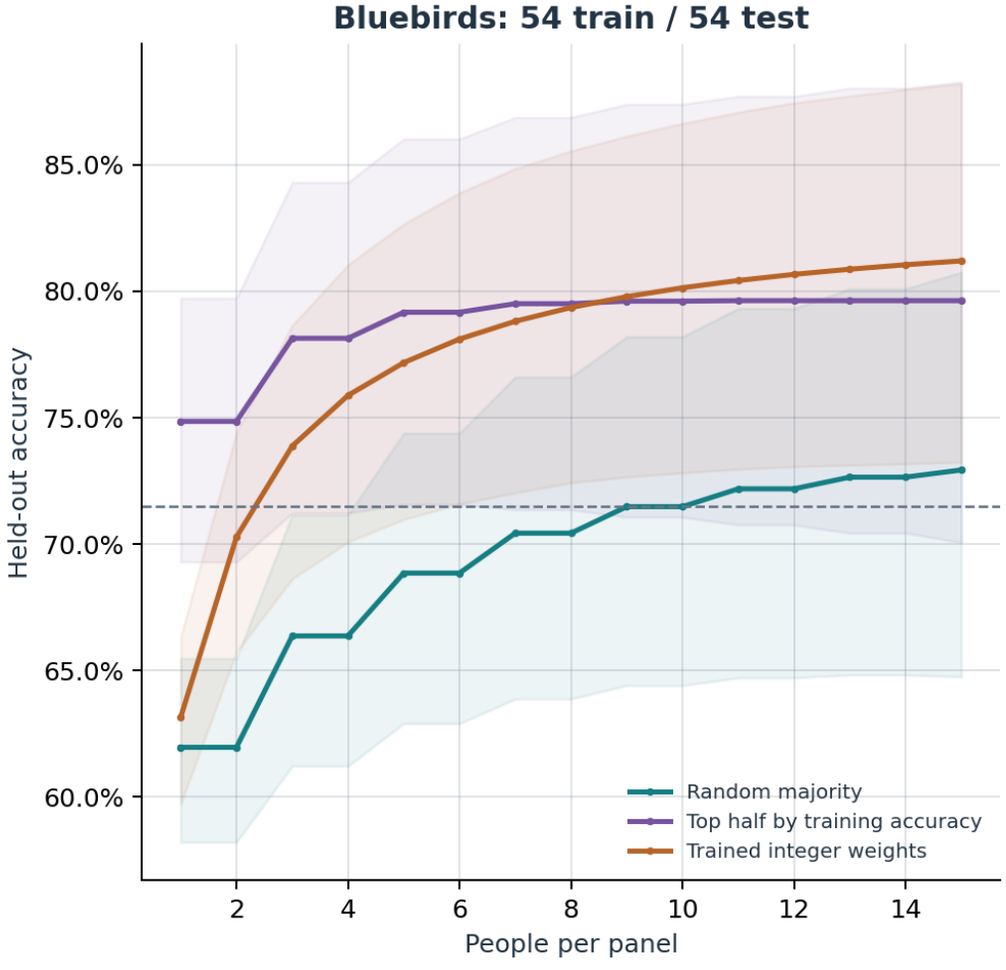
RETROSPECTIVE DIAGNOSTIC | Top: fraction of available votes that are correct. Bottom: exact benefit of nine versus one.

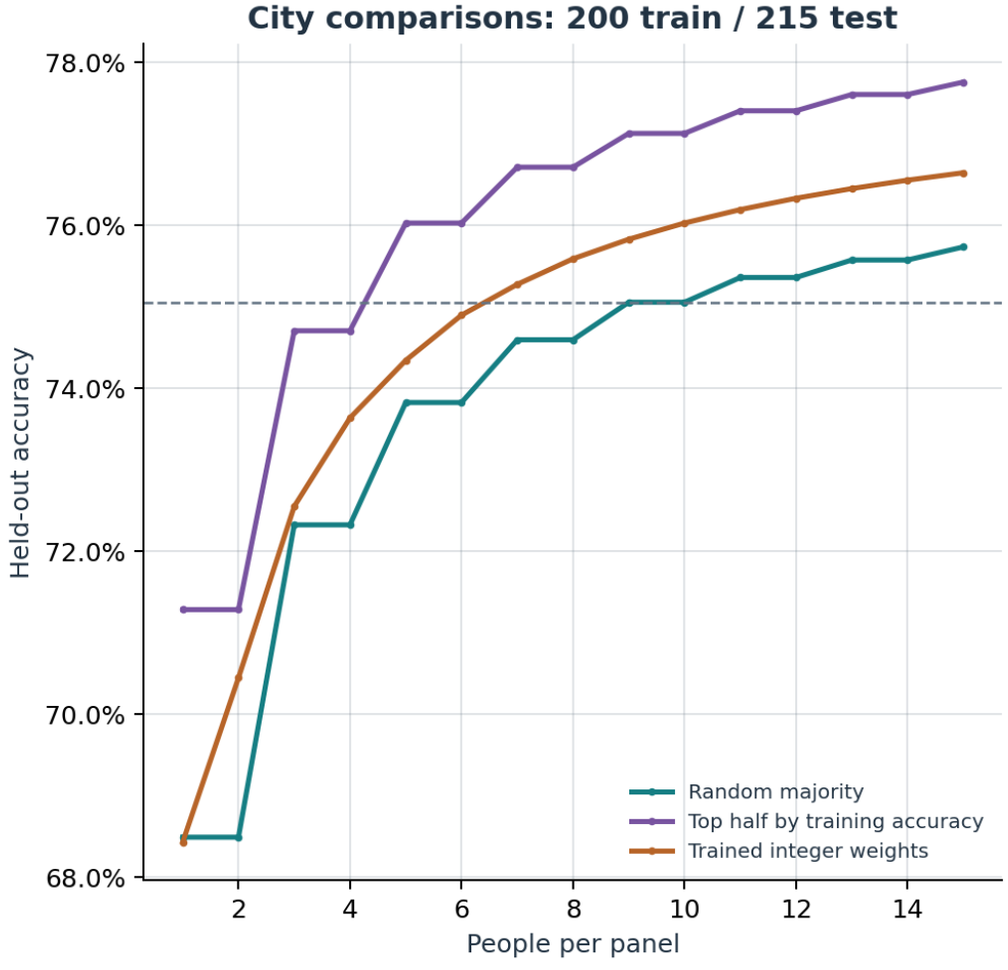


Each point uses recorded responses on one item; positive means improvement at k=9. These groups use known outcomes.
The relationship is partly mathematical: majority voting amplifies an observed pool leaning right or wrong. It is not a prospective skill test.

5. Expertise versus headcount on held-out questions

RECORDED RESPONSES | Selection and weights use training questions only. Every method is scored on the same held-out items.
The target is the held-out accuracy of nine random raters from that dataset.

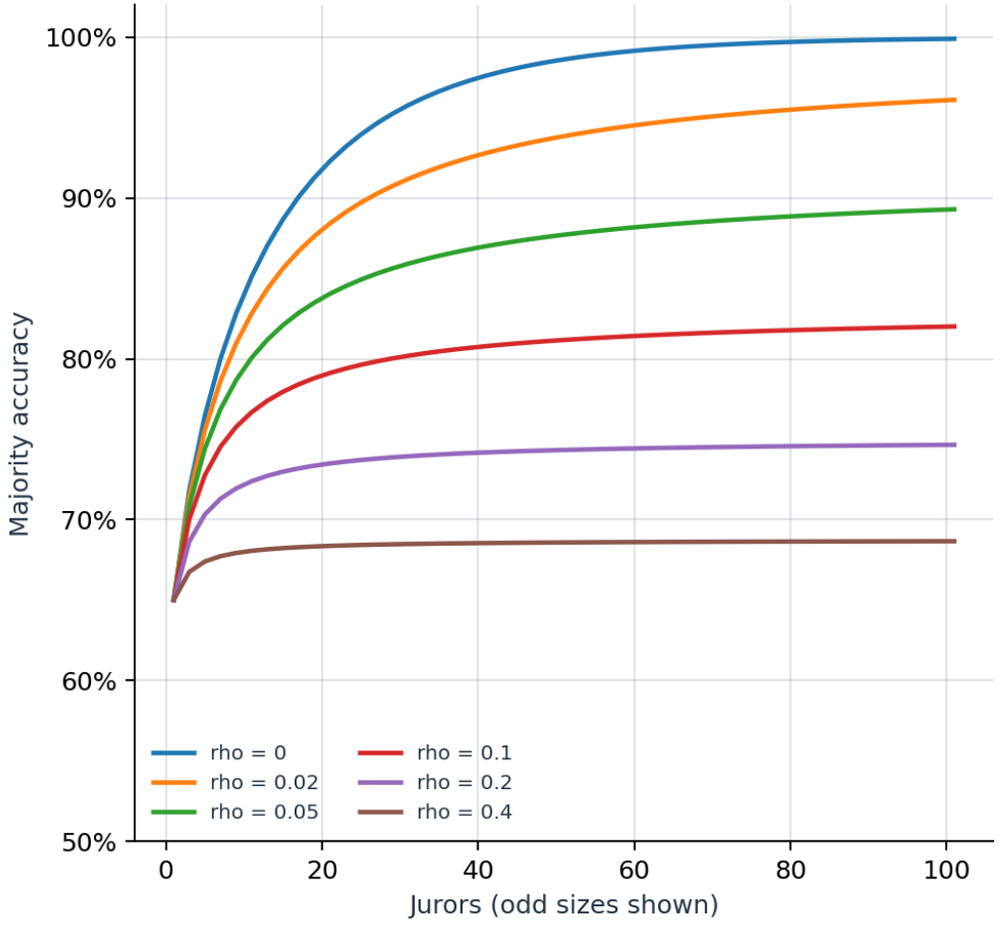


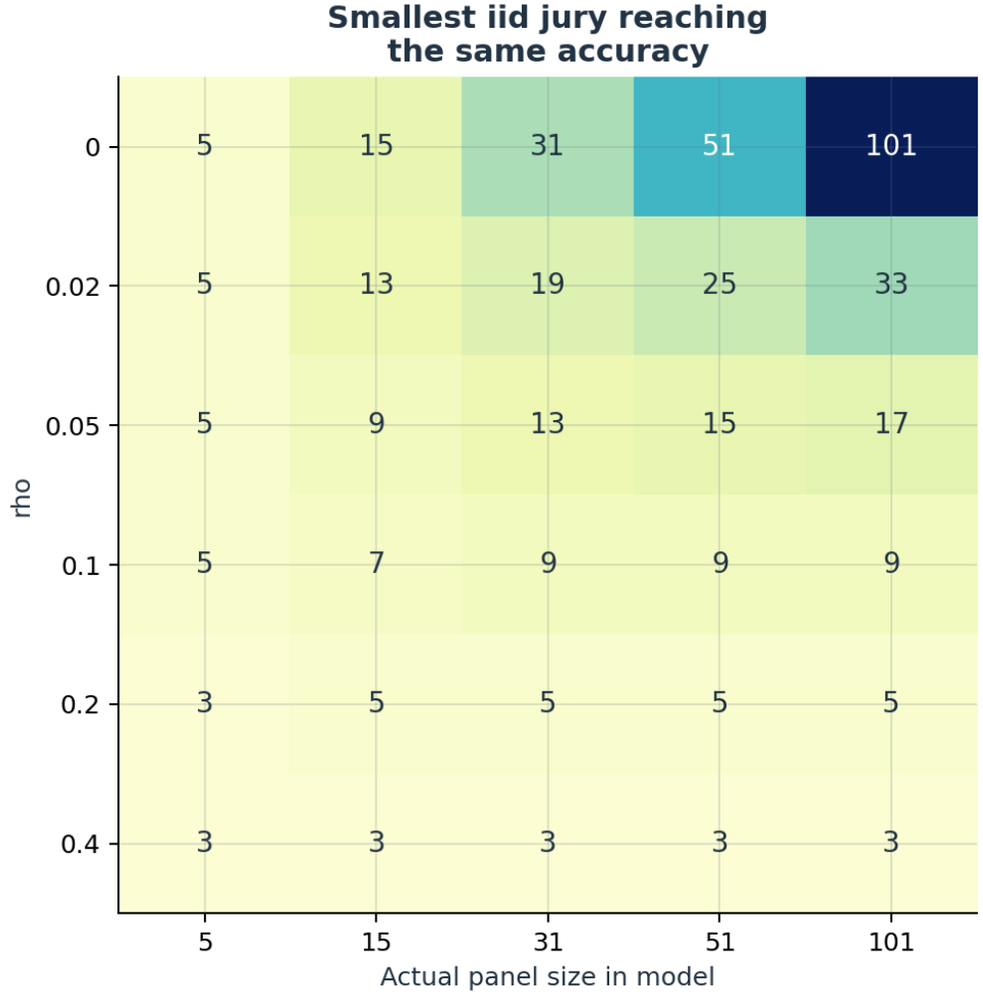


Exact subset scores for all methods; trained weights are discretized nonnegative log odds, not a claim of optimality.
Bird bands condition on fixed training. City split has disjoint city identities and uses complete items; no city confidence bands.

6. Shared case conditions can make accuracy plateau

THEORETICAL MODEL | Beta-binomial correctness: jurors share a latent item-specific success probability.
Mean individual accuracy remains 65%; rho denotes marginal correctness correlation.

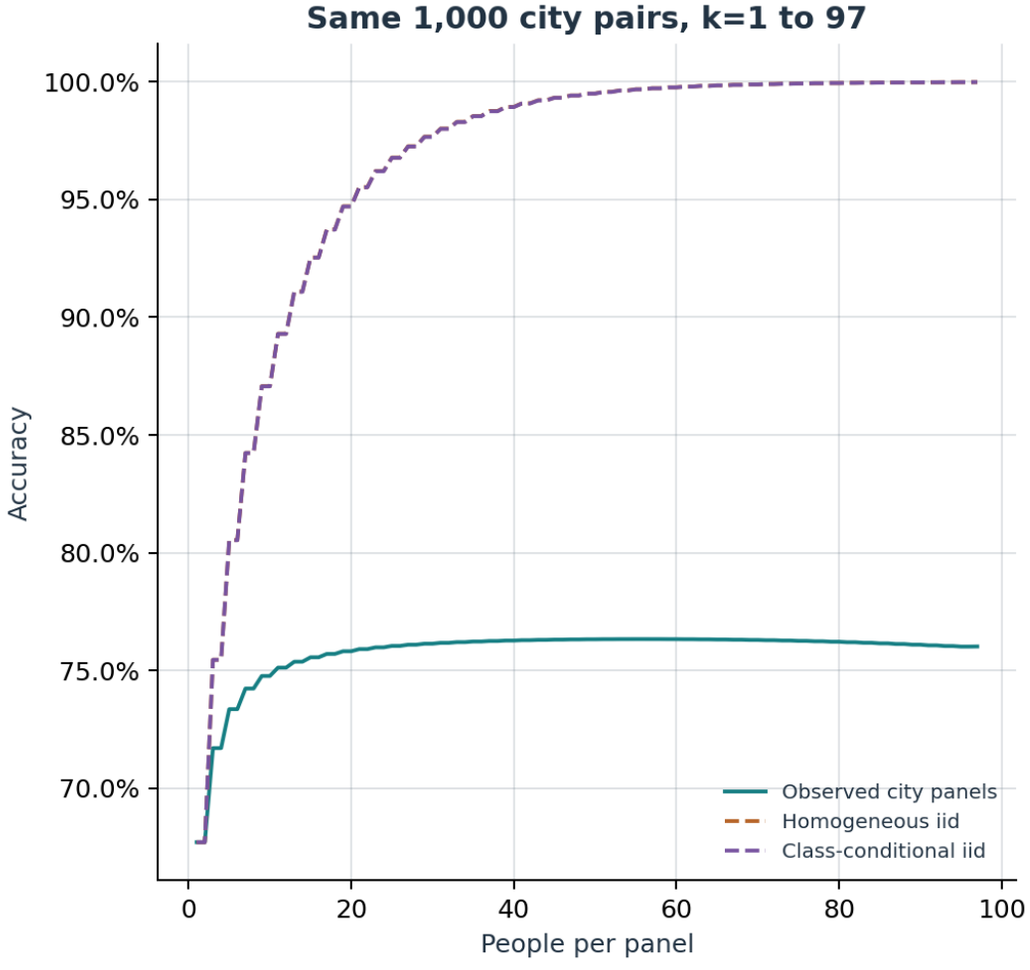


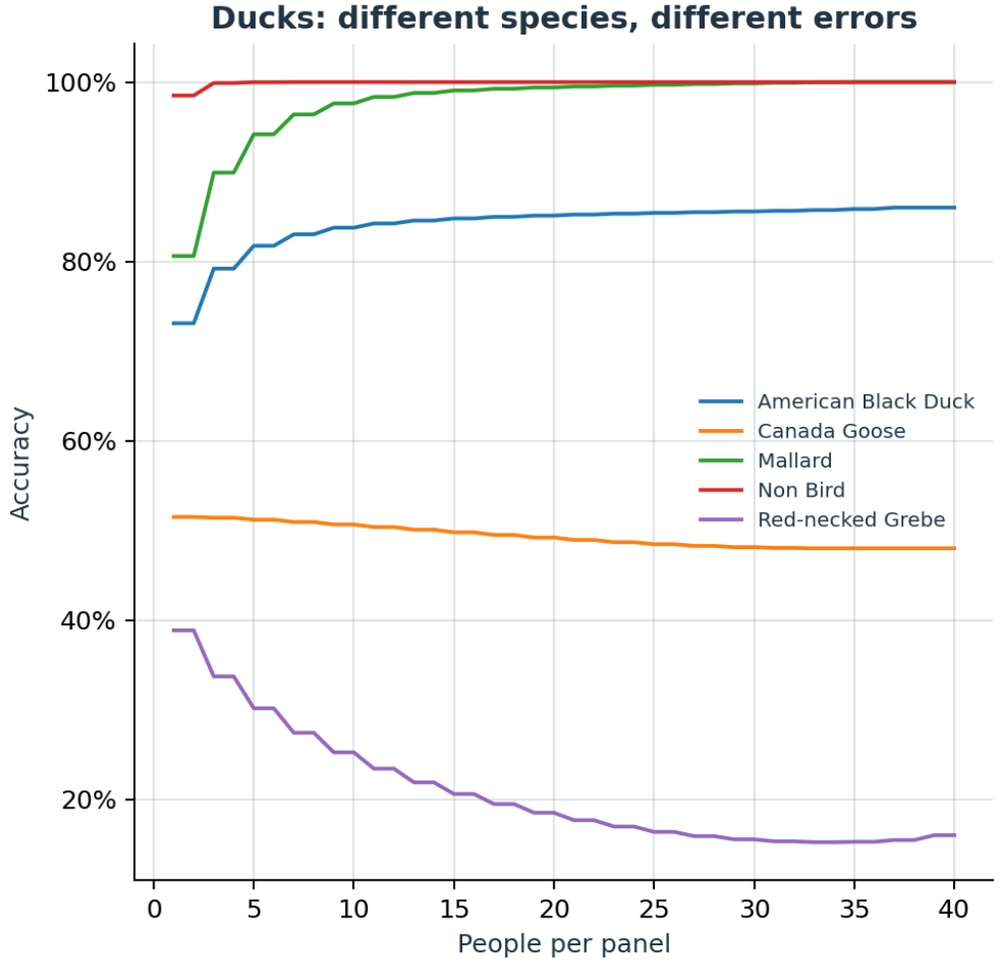


Every value on this page is analytical, not fitted to the datasets. A specified joint model is essential: pairwise correlation alone is insufficient.
This mechanism need not involve communication. The variance shortcut k / [1 + (k - 1) rho] is not used as accuracy equivalence.

Appendix. Full support and class-specific performance

RECORDED RESPONSES + CLASS-CONDITIONAL MODEL | Use the full city range and inspect the five Ducks species.





All curves average over fixed observed items. Every city item has at least 97 votes; every Ducks image has 40.
Class-specific curves for all datasets and matched class-conditional model curves are included in the accompanying CSV tables.

Methods, sources and limits

Accuracy and equivalence. A k-person panel is a uniform subset of distinct available respondents on an item. Binary majority ties receive half credit. Hypergeometric tails give exact expected accuracy; whole-count equivalence is the first supported crossing. Fractional values interpolate every integer size, retaining even/odd plateaus. A >K result is censored at the supported range, not an estimate of K+1.

Training. Bluebirds uses a seeded, truth-stratified half split of images. Cities uses complete 99-person items and disjoint sets of city identities; pairs crossing those sets are excluded. Rater selection keeps the top half by smoothed training accuracy. Weights are nonnegative, rounded log odds with a Beta(2,2) prior, capped at 10. All weighted subset scores are exact. These conventions were fixed before evaluating test outcomes.

Uncertainty. 4,000 stratified bird-image resamples and the existing 10,000 NBA wave resamples. The iid benchmark is refitted in every draw. Learned-policy bird bands condition on the fixed training result. City curves have no confidence intervals. This does not measure uncertainty from recruiting new people or establish causal effects of panel composition.

Data handling. Bluebirds: 108 images, 39 raters. Ducks: 240 images, 53 raters, 40 votes per image. Cities: 99 retained people, 1,000 pairs, 98,787 valid votes; 213 absent decisions encoded as incorrect are excluded. NBA: 465 games, 113 people, 9,788 complete records. Source hashes and all scoring tables accompany the report.

Sources. Rutchick et al. (2020), doi.org/10.1186/s41235-020-00256-z; NBA release osf.io/2rzcs/. Welinder et al. (2010), The Multidimensional Wisdom of Crowds; github.com/welinder/cubam (pinned commit in data_audit.json). Yu et al. city study release: osf.io/cuzqm/, file osf.io/vyx8a; Kurvers et al. reanalysis: doi.org/10.1126/sciadv.aaw9011.

Framework. Resnick et al., arxiv.org/abs/2106.01254v3 (rater equivalence); Dietrich and Spiekermann, Epistemic democracy with defensible premises (2013); Ladha, doi.org/10.1016/0167-2681(94)00068-P (correlated voting). This report adapts the human-label framework to known outcomes; it does not substitute human consensus for truth.